

Understanding Performance Bottlenecks of Large-Scale Agent-Based Simulations

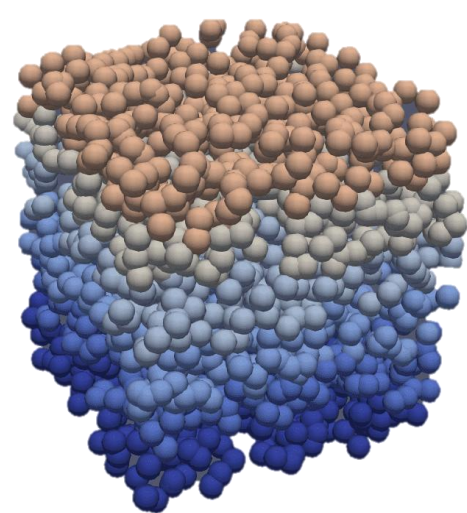
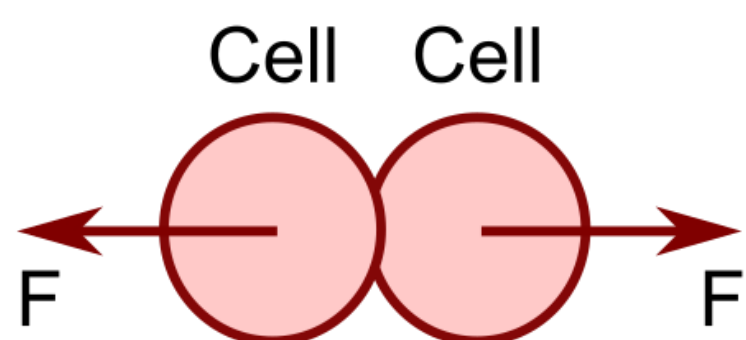
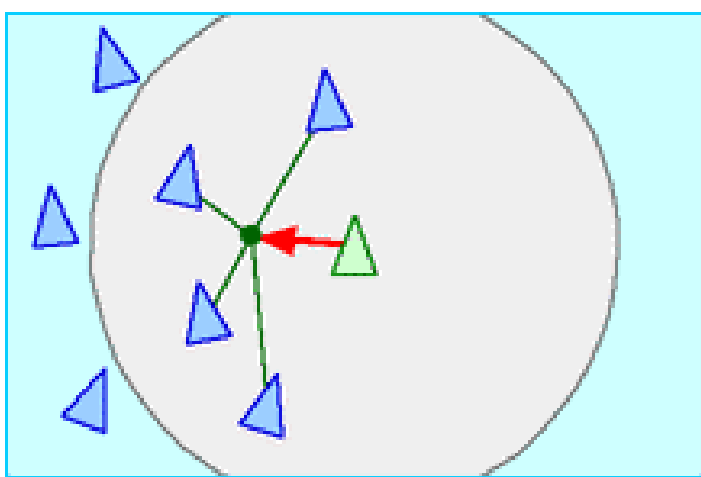
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Motivation

Agent-based simulations are increasingly used to simulate complex real-world phenomena in diverse scientific fields, yet their hardware performance characteristics remain poorly understood. As simulations scale to multi-billions of agents, data movement between compute units and memory becomes a critical bottleneck. **No prior work rigorously investigates these bottlenecks**, leaving a gap in our understanding of what limits scalability.

Introduction

- Agent-based simulations **model** real-world as populations of **autonomous agents** interacting in an environment. Their memory access patterns *resemble graph workloads'* pointer-chasing traversal, but with dynamic, spatially-evolving neighbor structure.
- We evaluate performance bottlenecks in agent-based simulations by profiling **BioDynaMo's** epidemiology use case on **Google Cloud C4 VMs** across agent counts from **10 million to 1 billion**.



Methodology

Table 1: Google Cloud C4 VM Specs

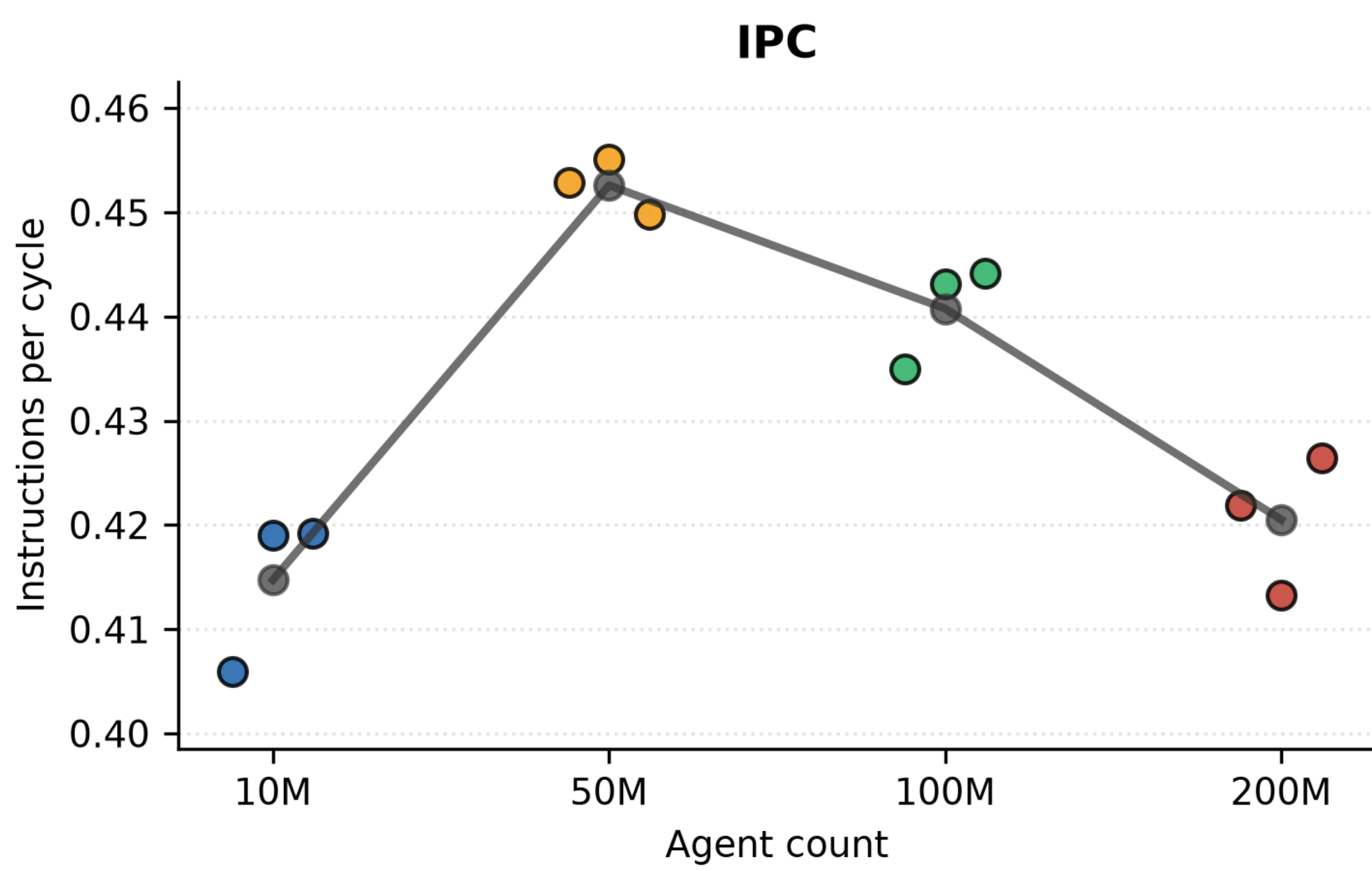
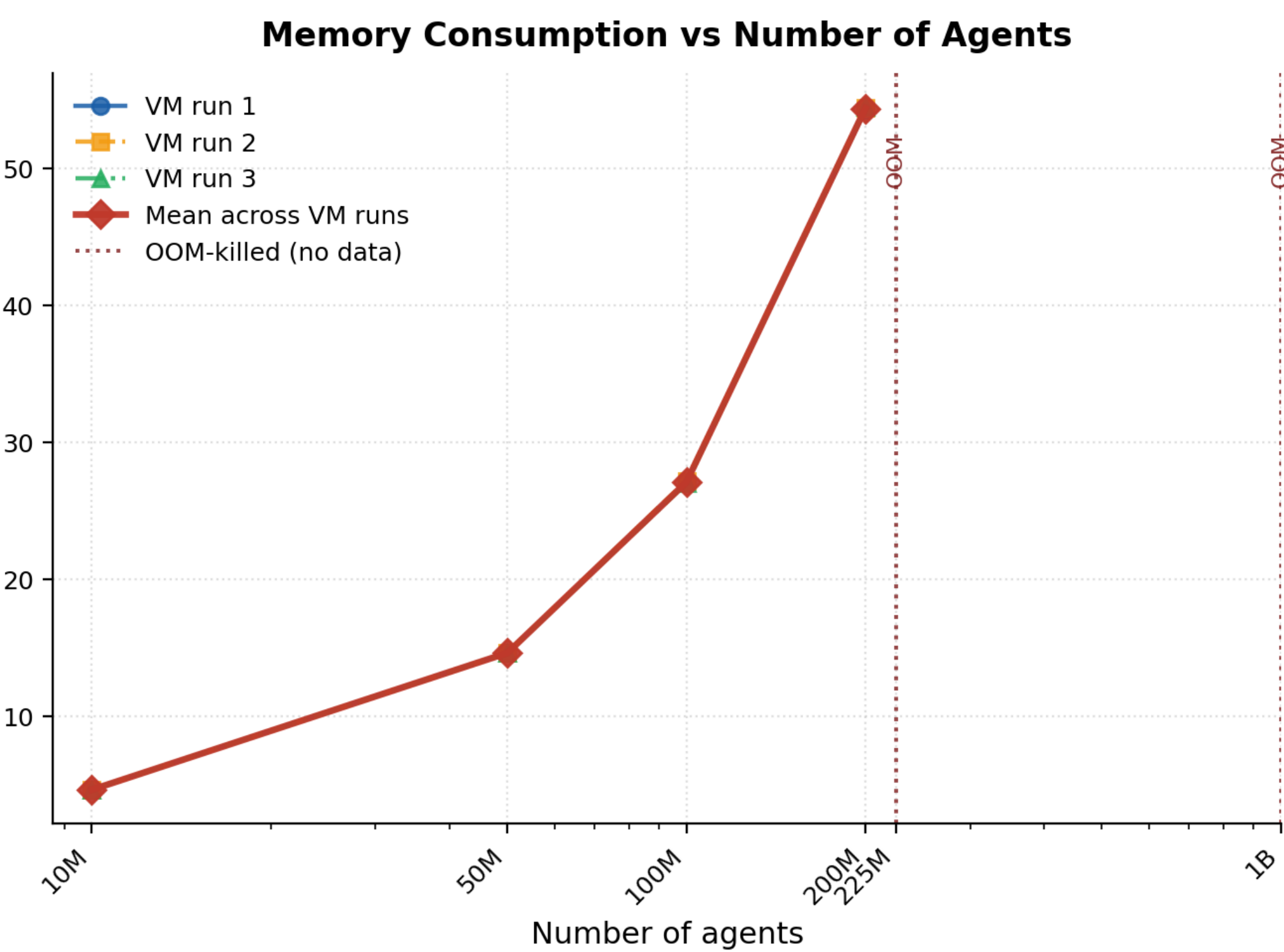
Processor	16-core Intel Emerald Rapids
Memory	64GB with four NUMA domains
OS	Ubuntu 22.04 LTS

- 3 runs per agent count x 3 individual VMs in parallel
- PMU counters collected via `perf` sampled at **1 Hz**
- Ran agent configurations at **16, 8, 4, 2 and 1 threads** to isolate compute parallelism from scale effects

Table 2: PMU Statistics

PMU events	Description	Metric
Instructions	Number of instructions	IPC
Cycles	Number of CPU cycles	IPC
MEM_LOAD_RETIRED.L1_MISS	Number of retired instructions that cause an L1 miss	L2/L1 miss ratio
MEM_LOAD_RETIRED.L2_MISS	Number of retired instructions that cause an L2 miss	L2 MPKI, L2/L1 miss ratio
TOPDOWN.MEMORY_BOUND_SLOTS	Pipeline slots that are memory bound	Mem-bound fraction
TOPDOWN.SLOTS_P	Total number of slots in the processor pipeline	Mem-bound fraction

Results and Analysis

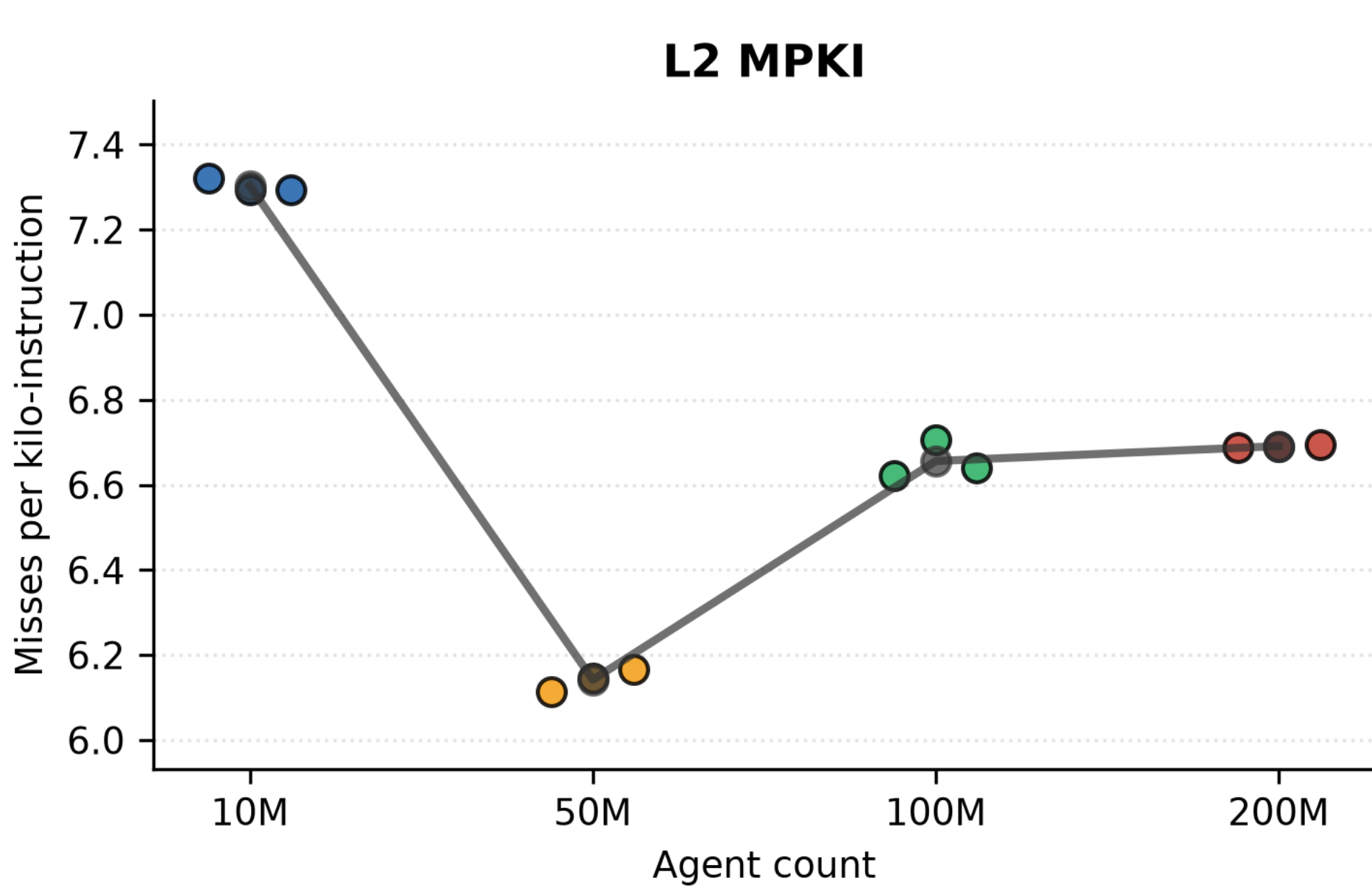
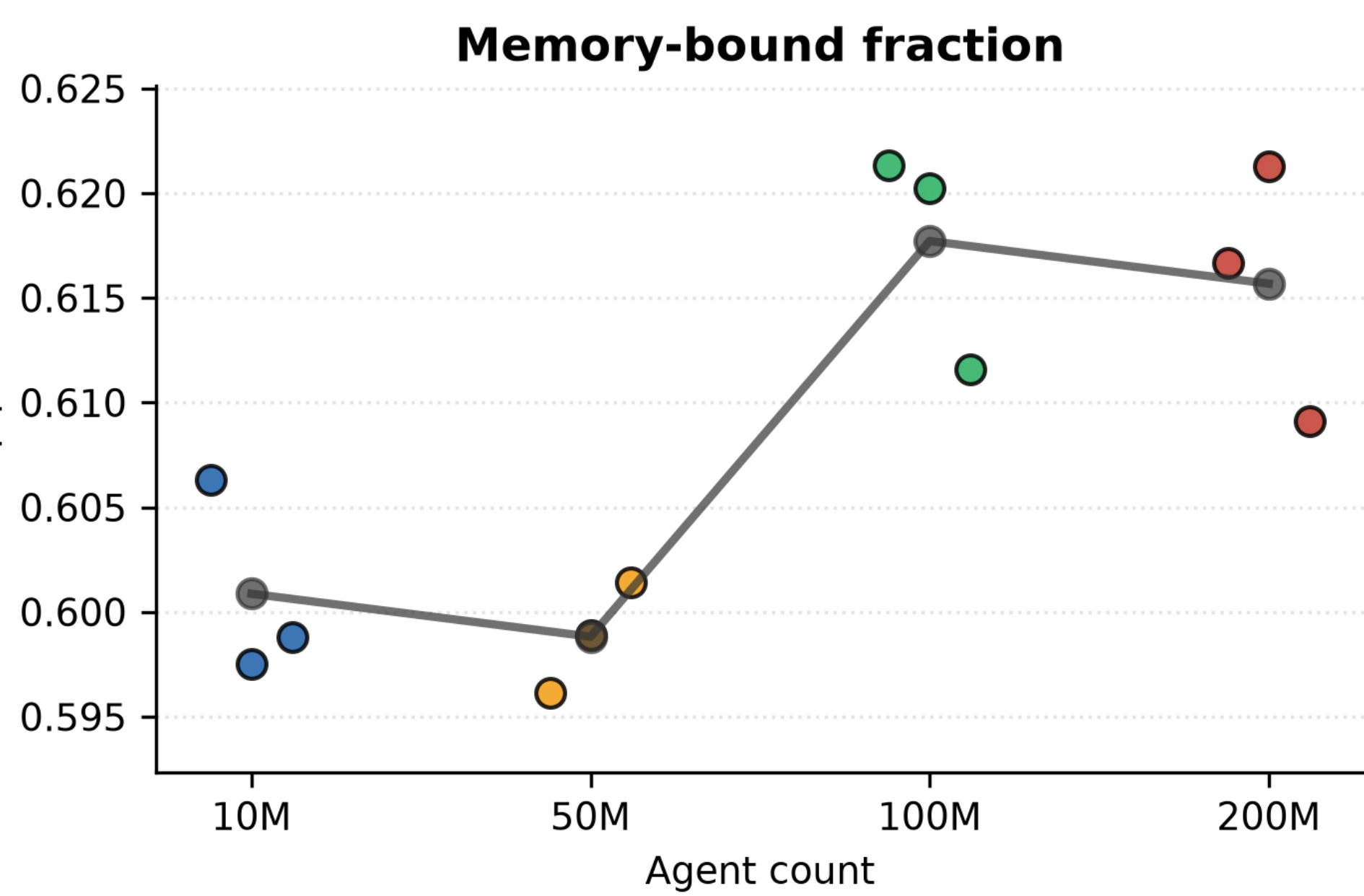


IPC and L2 MPKI:

- 1) **IPC remains consistently low** across all runs with a value of ~ 0.4 on a 16-core system, **dropping** from ~ 0.65 at 1 thread **as thread count increases**
- 2) L2 MPKI is non-monotonic across scale

Memory-bound fraction:

- 1) memory-bound fraction shows that the simulation is **memory-bound for over 60% of pipeline slots** across all agent counts



Memory consumption grows super-linearly with agent count

The 200M agent simulation consumed approximately ≈ 290 bytes per agent $\Rightarrow$ 200M agents is the practical upper bound for this VM configuration.

- Despite moderate cache MPKI, **performance is bottlenecked by memory accesses**;
- These metrics place this simulation in the **memory-bound** region of the DAMOV classification tree;
- Scaling agent-based simulations toward real-world population sizes requires **addressing data movement**, not just adding compute

Future works:

- Test on **bare metal** machines to rule out virtualization;
- Characterize temporal and spatial locality** for a precise DAMOV classification;
- Extend** our experiments to **architectures designed for highly-parallel execution** (e.g., GPUs, PuD)